

Practical - 25,26

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Class : SYIT	Batch : 1
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PRACTICAL - 25

AIM: Handle missing values in a dataset. Replace missing values with the column mean.

INPUT:

```
object PRACTICAL extends App {  
  
    val sales:List[java.lang.Double] = List(100.0, 200.0, null, 300.0, null)  
  
    val v = sales.filter(_ != null)  
  
    val mean = v.map(_.toDouble).sum / v.length  
  
    println("Mean Sales: " + mean)  
  
    val n = sales.map { value =>  
        if (value == null) mean  
        else value  
    }  
    println("Original Data:")  
    println(sales.mkString(", "))  
  
    println("After Replacing Missing Values:")  
    println(n.mkString(", "))  
  
}
```

OUTPUT:

```
Run PRACTICAL x
C:\Users\jiawa\.jdk\ms-17.0.20.1\bin\java.exe
Mean Sales: 200.0
Original Data:
100.0, 200.0, null, 300.0, null
After Replacing Missing Values:
100.0, 200.0, 200.0, 300.0, 200.0
Process finished with exit code 0
```

PRACTICAL - 26

AIM: Perform logistic regression using Breeze. Classify a dataset with binary labels.

INPUT:

```
import breeze.linalg._
object PRACTICAL extends App {

    val x = DenseVector(1.0, 2.0, 3.0, 4.0, 5.0) // Dataset
    val y = DenseVector(0.0, 0.0, 0.0, 1.0, 1.0) // Binary labels

    // Initial weight and bias
    var weight = 0.0
    var bias = 0.0

    // Sigmoid function
    def sigmoid(z: Double): Double = {
        1.0 / (1.0 + math.exp(-z))
    }

    for (i <- 1 to 1000) {
        for (j <- 0 until x.length) {
            val prediction = sigmoid(weight * x(j) + bias)
            val error = prediction - y(j)
            weight = weight - 0.01 * error * x(j)
            bias = bias - 0.01 * error
        }
    }
}
```

```
println("Predictions:")
for (i <- 0 until x.length) {
    val probability = sigmoid(weight * x(i) + bias)
    val result = if (probability >= 0.5) 1 else 0
    println("Input = " + x(i) + ", Probability = " + probability + ", Class = " + result)
}
```

OUTPUT:

```
Run PRACTICAL x
C:\Users\jiawa\.jdk\ms-17.0.20.1\bin\java.exe ...
Predictions:
Input = 1.0, Probability = 0.0515501057576029, Class = 0
Input = 2.0, Probability = 0.16412826954644827, Class = 0
Input = 3.0, Probability = 0.4149888799701166, Class = 0
Input = 4.0, Probability = 0.7193154698480789, Class = 1
Input = 5.0, Probability = 0.9025176083649726, Class = 1

Process finished with exit code 0
```